Paper 20: Harmonic Mood Palettes & Dirty Gradient Shading

Pillar: Graph-Loop Creative Pipelines — Autonomous Research Ecosystem

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Abstract

We hypothesize that Harmonic Mood Palettes & Dirty Gradient Shading achieves Strict ASE/ACO match on Python / OpenCV while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers. We present a formal mathematical model, a production implementation in `Python / OpenCV`, and an automated empirical verification harness targeting **Strict ASE/ACO match**. All benchmarks and invariants are reproducibly verified via our continuous Research Graph-Loop pipeline [3, 6, 2].

1 Introduction & Formal Hypothesis

Modern autonomous agent architectures face fundamental bottlenecks across constrained edge silicon, deterministic multi-modal synthesis, and decentralized economic clearing. In this work (Paper 20 within the *Graph-Loop Creative Pipelines* pillar), we address these limitations directly.

Core Hypothesis: We hypothesize that Harmonic Mood Palettes & Dirty Gradient Shading achieves Strict ASE/ACO match on Python / OpenCV while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers.

2 Mathematical Formulation

We formulate our system objective and state transition invariant as:

$$\mathcal{L}_{\text{total}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\left[\mathcal{D}_{\text{KL}}(P_{\text{target}} \parallel P_{\text{draft}})\right] + \lambda\|\Delta_{\text{state}}\|_2^2 \tag{1}$$

Subject to strict runtime verification bounds (`Strict ASE/ACO match`) executed across our hybrid compute fabric (Local Edge Silicon, Homelab Silo Server, and FAU Research Computing HPC nodes) [4, 5, 1].

3 System Architecture & Implementation

The reference architecture is implemented in `Python / OpenCV` across the following work package modules:

- `python/color_lab/dirty_gradient.py`
- `research_ecosystem/pillar2_creative.py`

4 Empirical Evaluation & Benchmarks

We executed the automated verification suite for Paper 20 and recorded the following empirical metrics (Table 1):

Table 1: Empirical Verification Results for Paper 20

Metric / Invariant	Measured Value
palette_size	3
strict_swatch_match	True
passed	True

5 Conclusion

We have demonstrated that *Harmonic Mood Palettes & Dirty Gradient Shading* satisfies its target specification (`Strict ASE/ACO match`) with deterministic reproducibility.

References

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